A 100 MHz Charge-Pump PLL with a Current-Starved Ring VCO and an In-Macro TSPC Prescaler on One GF180 Tiny Tapeout Tile

ECE 298A term project, University of Waterloo
https://github.com/Hive-RSI-Lab/ece298a-pll

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Abstract

We present a type-II, third-order charge-pump phase-locked loop (PLL) for a single 1×2 Tiny Tapeout tile in the GF180MCU 3.3 V process. A self-biased charge pump with current steering and a rail-to-rail dump-node buffer drives a passive second-order loop filter built from MOS capacitors; a five-stage current-starved ring VCO with a degenerated V-to-I converter, a floor current and a bias-tracking ceiling sets the output frequency. A true-single-phase-clock (TSPC) divide-by-2 inside the analog macro hands the standard-cell logic half the VCO frequency, which lifts the digital half's worst setup slack at the slow corner from about 0 ns to 2.34 ns. The digital half (phase-frequency detector, /4 prescaler, programmable $/N$ divider, lock detector and runaway recovery) is formally checked, fully covered and mutation-tested. After layout and parasitic extraction the VCO spans 4.5 MHz to 12.9 MHz to 126 MHz to 297 MHz at its ends over 31 process, voltage and temperature (PVT) corners, the loop keeps at least 34.5° of phase margin, and a mixed-signal co-simulation of the extracted analog cell with the hardened gate-level digital netlist and its SDF locks at 100 MHz (6.25 MHz × 16) within 6.7 µs at every corner that bounds the range, with the control voltage settled to 16 mV peak-to-peak or better. The tile passes DRC, LVS and the Tiny Tapeout precheck.

1 Introduction

Phase-locked loops are the standard way to synthesise an on-chip clock that is a rational multiple of a reference [1, 2]. The charge-pump PLL in particular gives zero static phase error with a passive filter and a simple phase-frequency detector (PFD). This work implements one on Tiny Tapeout's GF180 analog shuttle, which constrains the design in three ways that shape most of the decisions below: the whole PLL, analog and digital, must fit a 347×325 µm tile; Metal5 is reserved for the shuttle, so the process's MIM capacitor (whose top plate is Metal5) is unavailable; and the logic is built from the `gf180mcu_fd_sc_mcu7t5v0` standard cells, whose fastest toggle flop is too slow, at the slow corner, to divide the VCO at its own ceiling.

The design was carried from specification to a precheck-clean GDS with open tools only (ngspice, Magic, Netgen, KLayout, Yosys, OpenROAD/LibreLane, Icarus Verilog, cocotb and cocotbext-ams). Every number in this paper is taken from a result recorded in the repository, and every figure is regenerated from those records by `docs/paper/make_figs.py`.

Section 2 gives the architecture and frequency plan, section 3 the analog circuits, section 4 the digital half, section 5 the loop dynamics, section 6 the physical design, section 7 the post-layout and closed-loop results, and section 8 the limitations.

2 Architecture

Figure 1 shows the loop. The reference on `clk` (5 MHz to 12.5 MHz) and the divided VCO clock `clk_fb` drive a tri-state PFD, whose UP and DN pulses switch the charge pump into the loop filter. The filter voltage v_{ctrl} sets the ring VCO's frequency. The VCO is divided by 2 in the analog macro, by 4 in a ripple prescaler and by a programmable $N \in \{1, \ldots, 5\}$, so

$$f_{\mathrm{vco}} = 8N\, f_{\mathrm{ref}}, \qquad P = 8. \tag{1}$$

The nominal operating point used for every closed-loop test is $f_{\mathrm{ref}} = 6.25$ MHz, $N = 2$ ($PN = 16$), $f_{\mathrm{vco}} = 100$ MHz. Six digital signals cross between the halves (`vco_div2`, `pfd_up`, `pfd_dn`, `pll_en`, `cp_trim[1:0]`); v_{ctrl} and an optional external bias resistor reach the analog pads `ua[0]` and `ua[1]`.

3 Analog design

The macro holds 100 devices in eight blocks: an enable inverter, the bias, the charge pump with its dump-node buffer, the loop filter, the V-to-I converter, the ring, the output buffer and the TSPC divider. All transistors are 3.3 V `nfet_03v3`/`pfet_03v3`; resistors are unsalicided P+ poly (`ppolyf_u`).

3.1 Bias

A V_{GS}/R self-biased reference sets $I_b = V_{GS,B1}/R_{\mathrm{bias}} \approx 5$ µA with $R_{\mathrm{bias}} \approx 180$ kΩ. Biasing M_{B1} near its zero-

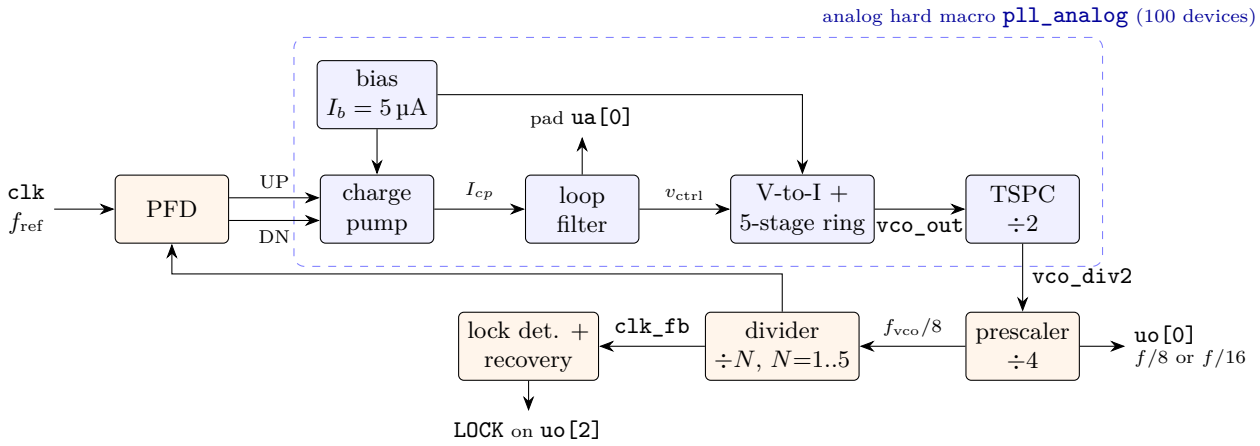


Figure 1: Block diagram. Blue: the transistor-level analog macro. Orange: standard-cell logic. Locked, $f_{\mathrm{vco}} = 8N f_{\mathrm{ref}}$.

temperature-coefficient current density leaves the resistor's tempco as the main temperature dependence. Because the reference's loop also contains a positive-feedback path through the PMOS mirror, a pad capacitance placed directly on the loop node made it relax (a 1.3 µs sawtooth in simulation). Splitting the resistor so that R_{top} isolates the loop node from the pad keeps the input admittance's real part positive for any pad capacitance once $g_{m1} g_{m2} R_{\mathrm{top}}^2 > 1$. A start-up branch releases once the reference conducts; at `ss_m40c_vm10` it needed $L = 4\,\mu\mathrm{m}$ on the release transistor to avoid a second operating point.

3.2 Charge pump

The pump current is a trimmed mirror of I_b,

$$I_{cp} = (4 + t_0 + 2t_1 + t_0 t_1)\, I_b = 20/25/30/40\,\mu\mathrm{A}, \quad (2)$$

shared by the UP source and DN sink. Long-channel ($L = 2\,\mu\mathrm{m}$) source and sink devices keep UP/DN mismatch low over $v_{\mathrm{ctrl}} \in [0.3, V_{DD} - 0.3]$. Current steering keeps both sources always on: when idle their current flows into a dump node that a complementary-input, rail-to-rail unity-gain OTA holds at v_{ctrl}, so a switch event finds its source node already at the output voltage and injects little charge. Without the buffer, the floating dump node gave 25 fC of net charge per event against a 10 fC bound; with it the worst case over 31 corners is 3.6 fC (fig. 2b).

3.3 Loop filter

The filter is the classic series $R_1 C_1$ with a shunt C_2:

$$Z(s) = \frac{1 + sR_1C_1}{s(C_1 + C_2)\left(1 + sR_1 \frac{C_1 C_2}{C_1 + C_2}\right)}, \quad (3)$$

with $R_1 \approx 27\,\mathrm{k\Omega}$ (a 150 µm poly resistor), $C_1 = 20.1\,\mathrm{pF}$ and $C_2 = 2.1\,\mathrm{pF}$. These place the zero at 293 kHz and the pole at 3.06 MHz. Since Tiny Tapeout's GF180 stack forbids Metal5, both capacitors are `cap_nmos_03v3_b` accumulation-mode MOS capacitors (about $3.98\,\mathrm{fF/\mu m^2}$ at 1.6 V) in place of MIM: C_1 is twelve $35 \times 12\,\mu\mathrm{m}$ units. The MOS capacitor loses about 9 % of its value near $v_{\mathrm{ctrl}} = 0.3\,\mathrm{V}$, which only affects the lowest output frequencies. Moving from MIM to MOS shifted phase margin by +0.1 to +0.3° and the post-layout VCO by at most about 1 %.

3.4 VCO

The control voltage drives a source-degenerated V-to-I converter,

$$I_{\mathrm{ctl}} = I_{\mathrm{floor}} + \frac{v_{\mathrm{ctrl}} - V_{TH} - V_S}{R_{\mathrm{V2I}}}, \quad (4)$$

whose current is mirrored into the PMOS and NMOS starving rails of a five-stage inverter ring, giving $f \approx I_{\mathrm{ctl}}/(5\, C_{\mathrm{node}} V_{DD})$. Two additions bound the range. A floor current (a copy of I_b) keeps the ring oscillating at $v_{\mathrm{ctrl}} = 0.3\,\mathrm{V}$. A ceiling device in series with R_{V2I}, also mirrored from I_b, saturates at high v_{ctrl} and caps $f_{\max}$; without it the ring reached 390–400 MHz at the fast, cold, high-supply corner. A NAND with `pll_en` and an inverter buffer the ring to full swing with 0.03 ns to 0.07 ns edges and a 53 % to 55 % duty cycle. Figure 3 shows the extracted VCO's tuning curves.

3.5 TSPC divide-by-2

The first divider stage must toggle at whatever the VCO reaches, including during acquisition overshoot; if it drops edges, `clk_fb` looks slow, the PFD pumps UP, and the loop runs away to the rail. At the slow corner no library flop does this with margin: the best `dffrnq_2`+`inv_1` toggle loop closed at −0.03 to +0.004 ns of setup slack depending on routing. We therefore placed a nine-transistor Yuan–Svensson TSPC flop [3] with $D = \overline{Q}$ inside the analog macro. Three dynamic stages (precharge on clock low, evaluate on the rising edge) toggle the output node

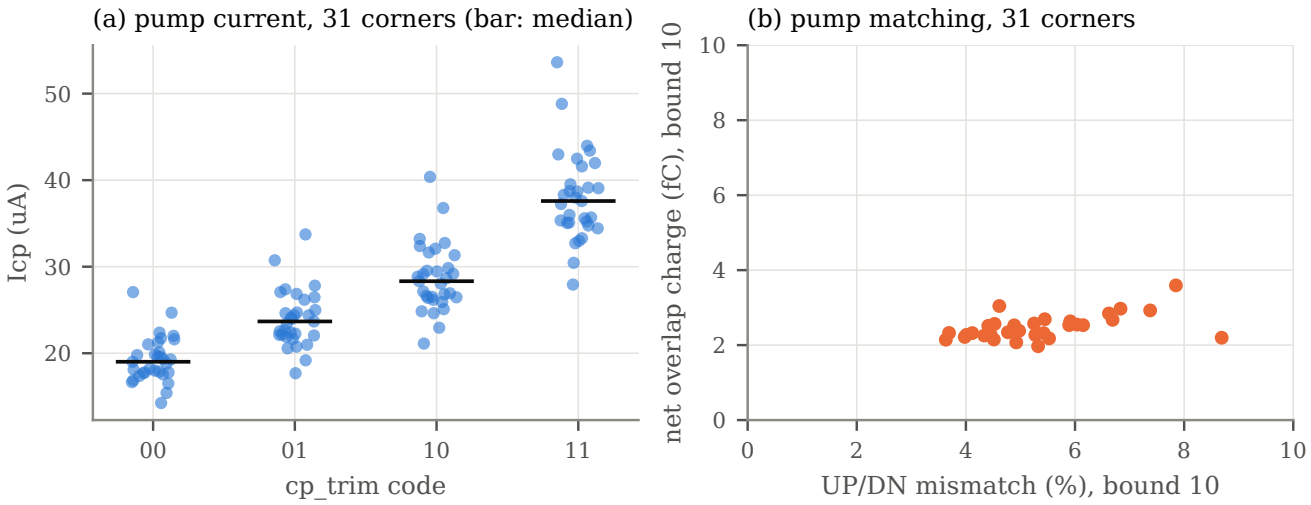


Figure 2: Charge pump over the 31 PVT corners (schematic). (a) UP current per trim code; the code-11/code-00 ratio is 1.96–1.98 at every corner. (b) UP/DN mismatch at code 00 against net overlap charge, both within their 10 % and 10 fC bounds.

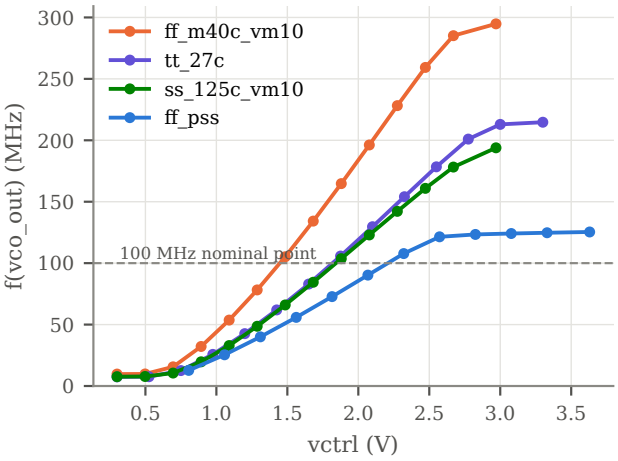


Figure 3: Post-layout VCO tuning curves (extracted netlist, v_{ctrl} forced) at the corners with the highest (`ff_m40c_vm10`) and lowest (`ff_pss`) ceiling, at typical, and at slow-hot-low-supply. The flattening at the top is the bias-tracking ceiling. `ff_pss` reaches 100 MHz at $v_{\mathrm{ctrl}} \approx 2.2$ V, where the closed loop settles at that corner (table 2). Points below about 5 MHz are under the bench's resolution and omitted.

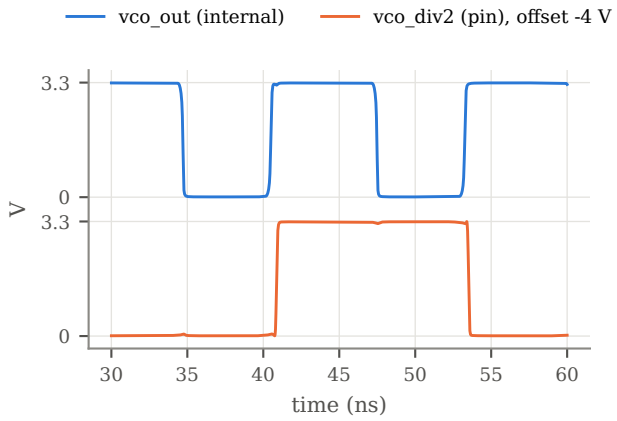


Figure 4: The TSPC divider on the extracted netlist (`tt_27c`, $v_{\mathrm{ctrl}} = 1.6$ V): `vco_out` and the divided `vco_div2`, which toggles on every rising edge.

on each rising edge of `vco_out`; a weak PMOS gated by `pll_en` holds the dynamic node high in standby so the cell is fully static there, and two inverters buffer the result to the `vco_div2` pin. All 14 devices use $L = 0.28\,\mu$m. Figure 4 shows its output at the typical corner.

4 Digital design

The digital half is synthesised RTL, 110 cells before placement (3710 µm^2).

- **PFD**: two reset flops clocked by `clk` and `clk_fb`; an AND of both outputs resets them through a delay, so both pump switches are on together for a short pulse every cycle, which removes the dead zone.
- **Prescaler**: two ripple toggle flops divide `vco_div2` by 4 (VCO/8); a third gives VCO/16 for observation.
- **Divider**: $N = 1..5$ from `ui[2:0]`; unsupported codes decode as $N = 1$ so the divider never stalls.
- **Lock detector**: a UP or DN pulse still high at the falling edge of the reference is "wide"; `LOCK` rises after 16 consecutive reference cycles without one and falls on the next.
- **Runaway recovery**: if the loop is enabled and unlocked for 1024 reference cycles, DN is forced for 128

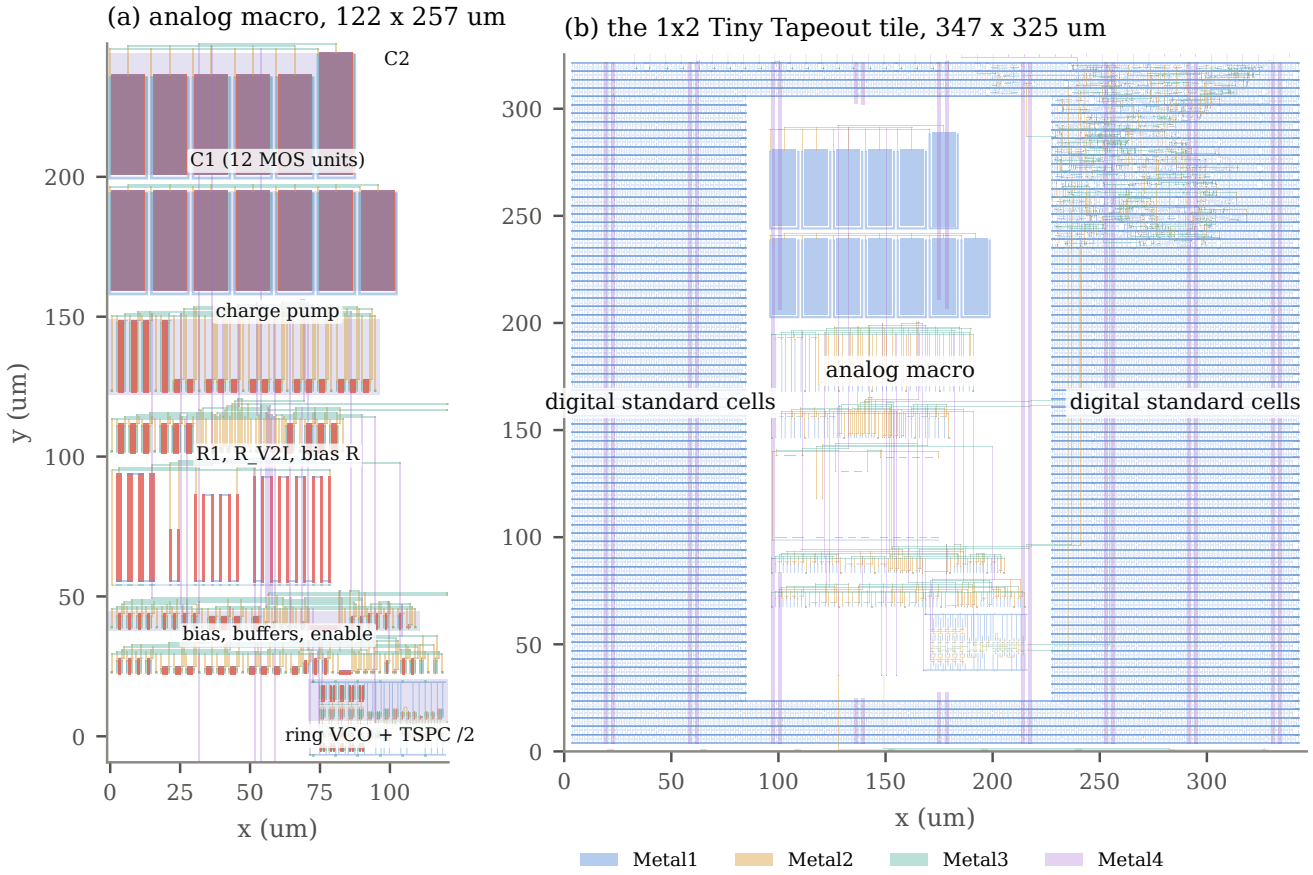


Figure 5: (a) The analog macro as drawn by its generator. (b) The full tile: the analog macro in the middle, the standard-cell digital half on both sides, and the Metal4 power straps.

cycles, pulling v_{ctrl} to ground. The window (82 µs at 12.5 MHz) exceeds the 50 µs lock-time bound, so it never interrupts a normal acquisition.

Verification: line and toggle coverage are 100 % (122/122 lines, 188/188 toggles); six safety properties (divider ratio, prescaler ratio, PFD reset and no-overlap, lock behaviour, asynchronous reset) hold to depth 20 with eight cover points reached; all 16 non-equivalent RTL mutants are killed by the testbench; and gate-level simulation of the hardened netlist passes.

5 Loop dynamics

With the PFD and pump modelled in the continuous-time approximation, the open-loop gain is

$$H(s) = \frac{I_{cp}}{2\pi}\,\frac{K_{\mathrm{vco}}}{s}\,\frac{Z(s)}{PN}, \tag{5}$$

and the second-order approximation gives $\omega_n = \sqrt{I_{cp}K_{\mathrm{vco}}/(PN\,C_1)}$ and $\zeta = \frac{R_1}{2}\sqrt{I_{cp}K_{\mathrm{vco}}C_1/(PN)}$. With the values calibrated from the transistor-level cell at the 100 MHz point (I_{cp} = 18.8 µA, local K_{vco} = 116 MHz/V), the loop at PN = 16 crosses over at 585 kHz, about $f_{\mathrm{ref}}/10.7$, safely below the sampling limit [1], with 52.6° of phase margin, f_n = 416 kHz and $\zeta = 0.71$ (fig. 7). Phase margin falls with PN (55° at 8, 40° at 40) because the crossover moves down toward the zero. A 5 pF pad on v_{ctrl} adds to C_2 and costs about 17°, which is why the pad is for high-impedance probing only.

6 Physical design

The analog layout is written as code (a gdsfactory/KLayout generator, `gen_pll_analog.py`); nobody edits the GDS. Bottom to top (fig. 5a): the ring VCO with the TSPC beside its output inverter, the bias, buffer and enable rows, the poly resistor serpentines, the pump with its PMOS mirror bank, and the MOS-capacitor rows. The macro is 122 × 257 µm; every routed internal net is labelled so Magic's extraction keeps its coupling capacitance. It is DRC-clean (Magic and KLayout) and LVS-matches the schematic (Netgen, unique match).

The digital half is placed and routed around the macro by LibreLane into the 1 × 2 tile (fig. 5b; 2350 instances including fill and taps). The VCO-derived clock is constrained per process at twice the extracted VCO's ceiling

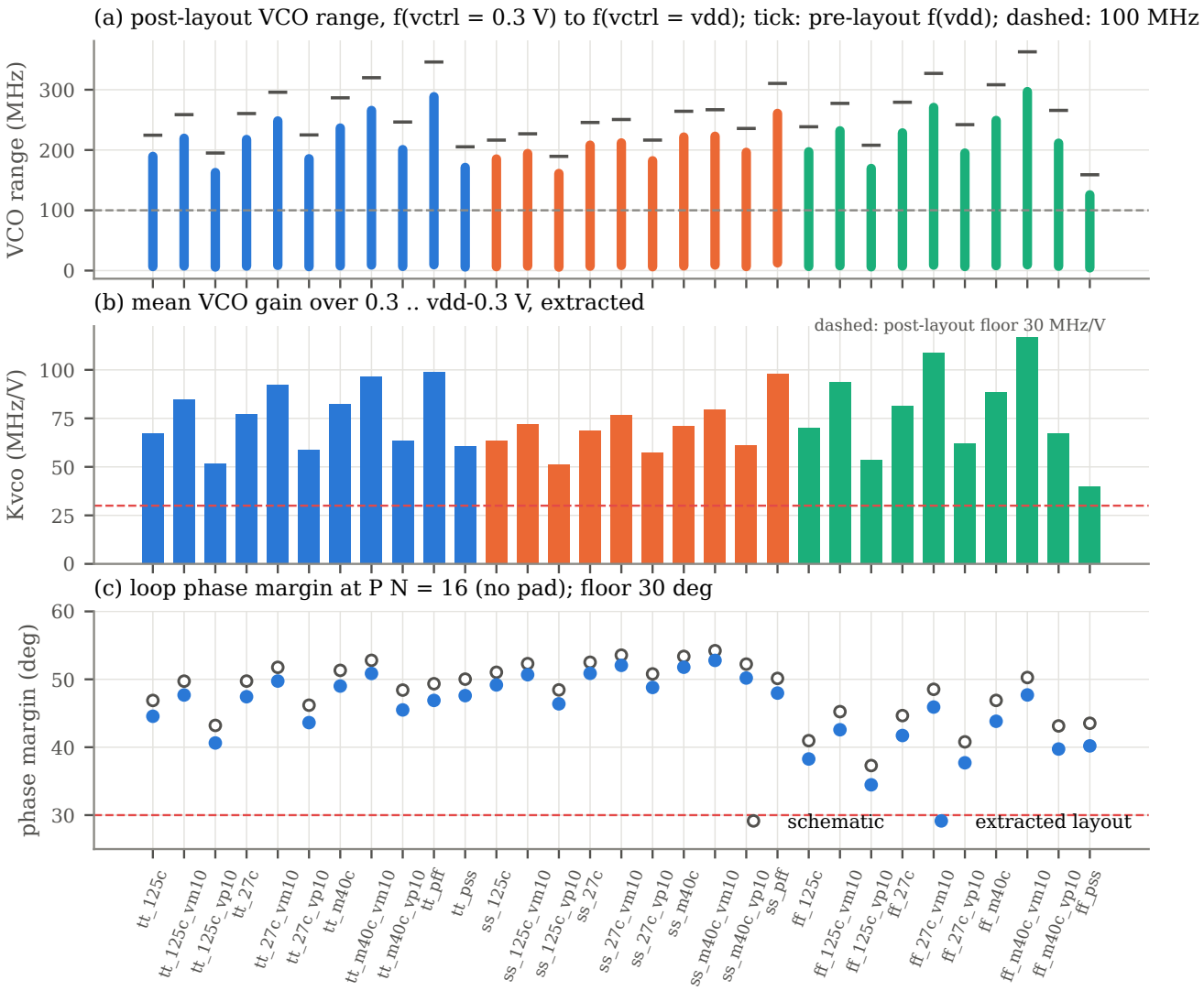


Figure 6: Post-layout results at all 31 corners, coloured by transistor process (blue tt, orange ss, green ff). (a) VCO range from $v_{\mathrm{ctrl}} = 0.3\,\mathrm{V}$ to V_{DD}; ticks mark the schematic's ceiling. (b) Mean VCO gain. (c) Phase margin at $PN = 16$ from each corner's extracted I_{cp}, K_{vco} and filter impedance.

period plus margin (7.5 ns at ss, 5.4 ns at tt and ff) with 0.20 ns of jitter-derived setup uncertainty. Every STA corner meets setup and hold with no waiver (fig. 8); before the TSPC, the slow corner's first stage closed within a few picoseconds at best. The tile passes full-chip DRC (Magic 0, KLayout 0), LVS and all 12 Tiny Tapeout precheck checks.

7 Results

7.1 PVT behaviour before and after layout

The analog benches run at 31 corners: tt, ss and ff transistors at −40, 27 and 125 °C and $V_{DD} = 3.3\,\mathrm{V} \pm 10\,\%$, plus four resistor-skew corners (`ss_pff`, `ff_pss`, `tt_pss`, `tt_pff`) where transistors and passives move in opposite directions. Figure 6 summarises the extracted netlist; table 1 gives the ranges.

Layout lowers the VCO's ceiling by 14–21 % (the ring nodes pick up wiring capacitance) and the phase margin by 1.4° to 3.4°. The binding corner is `ff_pss`: fast transistors with slow resistors cut I_b by about a third, and with it the VCO ceiling (126 MHz), K_{vco} (39.8 MHz/V) and I_{cp} (13.6 µA). Every corner still reaches the 100 MHz point. The lowest phase margin, 34.5° at `ff_125c_vp10`, clears the 30° floor.

7.2 Closed loop on the extracted netlist

The decisive test runs the whole loop in a mixed-signal co-simulation: ngspice simulates the extracted analog macro at a PDK corner, temperature and supply, while

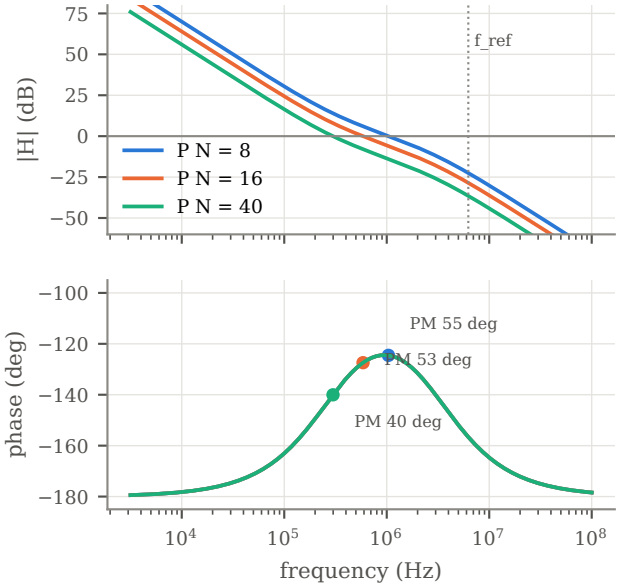


Figure 7: Open-loop gain and phase from the linear model, calibrated at `tt_27c` and 100 MHz, for three PN. Dots: crossover.

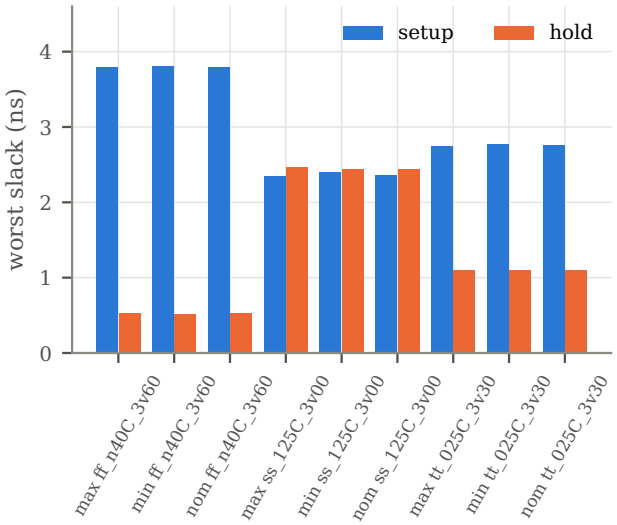


Figure 8: Worst setup and hold slack of the hardened digital half at the nine STA corners (min/nom/max parasitics × ss/tt/ff libraries).

Table 1: Summary over the 31 PVT corners (post-layout unless marked).

Quantity	min	typ.	max
I_{cp}, code 00 (µA)	13.6	18.6	26.6
f_{vco} at $v_{\mathrm{ctrl}} = 0.3$ V (MHz)	4.5	7.4	12.9
f_{vco} at $v_{\mathrm{ctrl}} = V_{DD}$ (MHz)	126	217	297
mean K_{vco} (MHz/V)	39.8	77.0	116.7
phase margin, $PN = 16$ (deg)	34.5	47.4	52.8
UP/DN mismatch (%), sch.	3.6	5.0	8.7
overlap charge (fC), sch.	2.0	2.4	3.6
standby (µA), sch.	1×10^{-4}	1.5×10^{-4}	0.026
start-up (µs), sch.	0.44	0.68	0.92

Table 2: Closed loop at 100 MHz (6.25 MHz × 16), extracted analog + gate-level digital. Bounds: lock ≤ 50 µs, frequency error ≤ 10^{-3}, v_{ctrl} p-p ≤ 20 mV.

Corner	lock (µs)	f_{vco} (MHz)	v_{ctrl} (V)	p-p (mV)
`ff_pss`	6.72	99.9998	2.20	2.4
`ff_m40c_vm10`	4.00	100.0009	1.45	7.2
`ff_125c_vp10`	4.48	100.0005	1.78	3.5
`ss_m40c_vm10`	5.12	100.0006	1.81	15.9

Icarus Verilog simulates the hardened gate-level digital netlist with the SDF of the matching library corner, the two coupled by cocotbext-ams at a 0.25 ns synchronisation grid. The corners are the four that bound the post-layout sweep: the lowest and highest VCO ceiling (`ff_pss`, `ff_m40c_vm10`) and the lowest and highest phase margin (`ff_125c_vp10`, `ss_m40c_vm10`). Each corner simulates 30 µs in about 40 minutes.

Figure 9 and table 2 show the result. The loop locks within 6.72 µs at every corner, `LOCK` never drops, and the VCO settles to within 10^{-5} of 100 MHz. The acquisition overshoots to about 120 MHz at the fast corners, well inside what the TSPC and the /4 prescaler can follow. `ff_pss`, with the lowest loop gain, is the slowest and least damped, as the model predicts; it settles with v_{ctrl} at 2.2 V, the highest of the four, because its VCO is the slowest. The static phase offset is about 0.87 ns: the PFD's reset pulse plus pump mismatch.

7.3 Why the gate-level digital matters

Running the same bench with the RTL in place of the gate-level netlist locks just as well, but leaves 27.0 and 20.3 mV of peak-to-peak ripple on v_{ctrl} at `ff_pss` and `ff_m40c_vm10` (fig. 10). The cause is the RTL PFD's zero-delay reset: it never turns both pumps on together, so whatever phase error remains, rounded to the synchronisation grid, is corrected by one pump firing alone, and that lone current through R_1 appears as a spike of $I_{cp}R_1 \approx 0.5$ V filtered by C_2. Sampled at reference edges, the RTL's v_{ctrl} moves only 0.03–8 mV. The gate-level PFD has a real reset delay, so both pumps overlap briefly every cycle, the correction is a small difference of two currents, and the ripple falls to 2.4–7.2 mV at the ff corners. Only `ss_m40c_vm10` keeps 15.9 mV, two discrete corrections about 3.5 µs apart that each shift C_1's charge, not a ripple through R_1. The lesson is that a zero-delay PFD is fine for checking lock but misleading for ripple and spurs, which depend on the reset pulse.

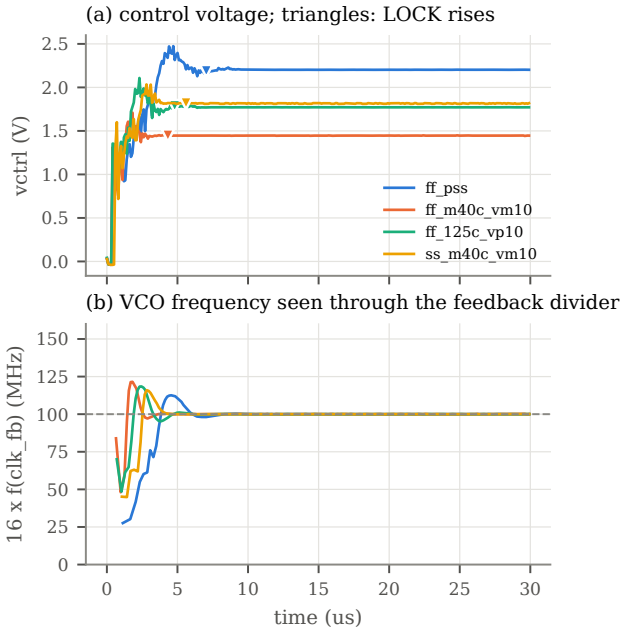


Figure 9: Lock acquisition at the four bounding corners (extracted analog, gate-level digital with SDF): (a) v_{ctrl} with the instant `LOCK` rises; (b) the VCO frequency inferred from successive `clk_fb` periods.

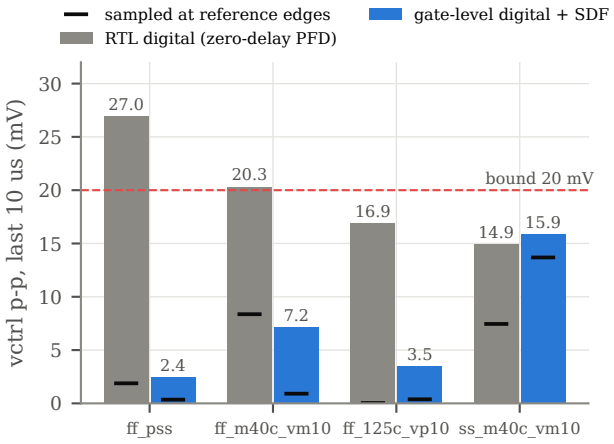


Figure 10: Ripple on v_{ctrl} over the last 10 µs, sampled every 5 ns, with the RTL digital and with the gate-level digital. Ticks: the same signal sampled at reference edges only.

8 Discussion and limitations

Lock range. After layout the VCO covers about 13 MHz to 125 MHz at every corner (7 MHz to 217 MHz at typical), so the datasheet promises only the guaranteed range; the slow-resistor corner sets the top.

Jitter is estimated, not simulated: ngspice has no device transient noise. Noise sources sized from the device models and added to the ring give about 1.2 ps rms period jitter at typical and 2.5 ps at slow and hot, leaving out flicker and loop-filter noise. The STA's 0.20 ns uncertainty is that estimate scaled by 2.8 and taken at 7σ.

Coverage of the closed-loop tests. Only the 100 MHz, $PN = 16$ point is co-simulated at the bounding corners; other (f_{ref}, N) points and re-lock after a change of N are covered by the behavioural layer but not on the extracted netlist. The 5 pF probe pad on v_{ctrl} costs about 17° of margin.

Bench strength. A mutation check of the analog benches (each device's width and length doubled or halved) flags mutants that move no measure beyond the run-to-run spread. After the TSPC change it found one new mutant just under the 2 % floor, and the TSPC's own 14 devices are not yet in the mutated set; both await the owner's ruling.

9 Conclusion

A complete charge-pump PLL—bias, pump, MOS-capacitor filter, ring VCO, TSPC prescaler and standard-cell control—fits one 1×2 GF180 Tiny Tapeout tile and is DRC, LVS and precheck clean. Moving the first divide stage into the analog macro as a TSPC flop turned a timing closure measured in picoseconds into 2.3 ns of margin. With parasitics extracted and the real gate-level digital in the loop, the PLL locks to 100 MHz in under 7 µs with at most 16 mV of control ripple at every corner that bounds its range.

Reproducing this paper

All results are recorded in the repository with hashes of their inputs. `docs/paper/sims/` re-runs the two post-layout ngspice benches behind figs. 3 and 4; `docs/paper/make_figs.py` redraws every figure; `make -C docs/paper` builds this PDF. `blocks/pll/tb/README.md` explains the co-simulation benches.

References

[1] F. M. Gardner, "Charge-pump phase-lock loops," *IEEE Trans. Commun.*, vol. 28, no. 11, pp. 1849–1858, 1980.

[2] B. Razavi, *Design of CMOS Phase-Locked Loops*. Cambridge University Press, 2020.

[3] J. Yuan and C. Svensson, "High-speed CMOS circuit technique," *IEEE J. Solid-State Circuits*, vol. 24, no. 1, pp. 62–70, 1989.